

Year 9

Test 1-Section 1



Baldivis
Secondary College

No resources allowed

Name: Marks

Class: _____

Time: 10 minutes

Total Mark(1): ____/16

Total Mark(1+2): ____/71

1. Write the following in index form

[3 marks]

a) $8 \times 8 \times 8 \times 8$

8^4 ✓

b) $f \times f \times f \times f \times f \times f$

f^6 ✓

c) $3 \times a \times a \times b \times b$

$3a^2b^2$ ✓

2. Evaluate the following:

[4 marks]

a) 5^2

25 ✓

b) 7^1

7 ✓

c) 12^0

1 ✓

d) 3^{-2}

$\frac{1}{9}$ ✓

3. Simplify the following expressions

[4 marks]

a) $3a + 6a$

$9a$ ✓

b) $12b - 7b$

$5b$ ✓

c) $4c \times 5d$

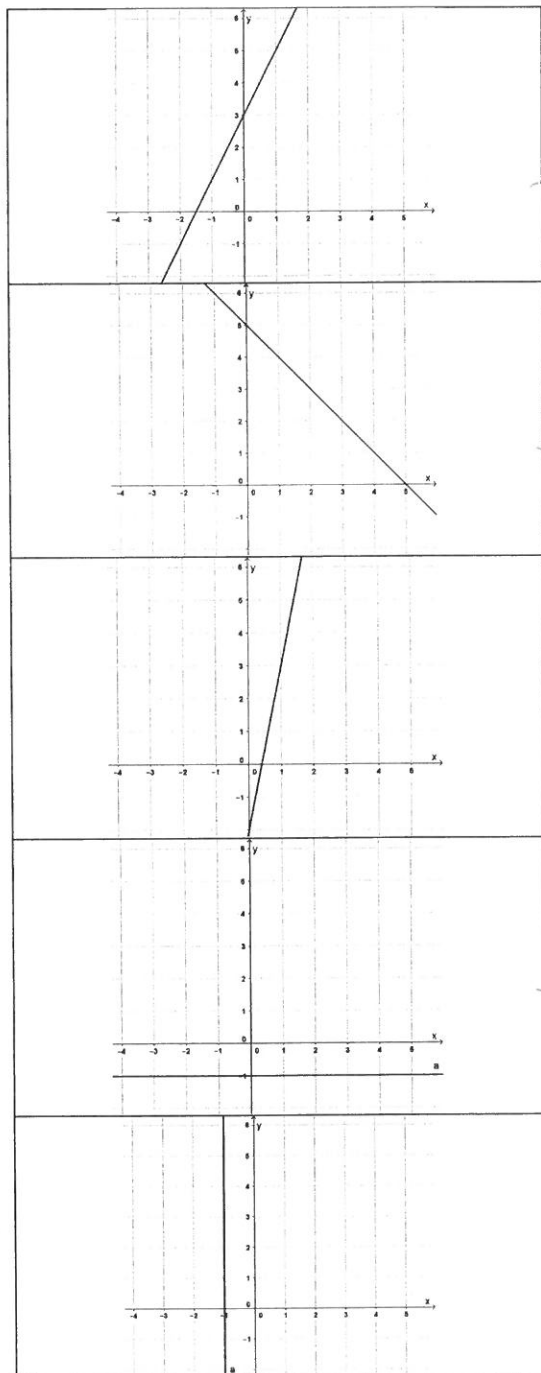
$20cd$ ✓

d) $\frac{8de}{4e}$

$2d$ ✓

4. Match the sketch with the rule

[5 marks]



$$y = -1$$

$$y = 2x + 3$$

$$x = -1$$

$$y = 5 - x$$

$$y = 5x - 2$$

Year 9

Test 1-Section 2



Baldivis
Secondary College

Resources allowed-Scientific calculator/One page of notes

Name: _____

Class: _____

Time: 45 minutes

Total Mark: ____/55

1. Use the Index Laws to simplify each of the following:

[13 marks]

a) $12^5 \times 12^3$ 12^8 ✓

b) $10^8 \times 10^{-5}$ 10^3 ✓

c) $x^7 \times x^9$ x^{16} ✓

d) $2^4 \times 4$ 2^6 ✓

e) $3^9 \div 3^3$ 3^6 ✓

f) $p^8 \div p^5$ p^3 ✓

g) $9t^5 \div 3t^{-3}$ $3t^8$ ✓✓

h) $(5^3)^4$ 5^{12} ✓

i) $(y^7)^2$ y^{14} ✓

j) $(2k^4)^3$ $8k^{12}$ ✓✓

2. Write with positive indices:

[3 marks]

a) 5^{-3} $\frac{1}{5^3}$ ✓

b) $\frac{2}{e^{-7}}$ $2e^7$ ✓

c) $\frac{7x^4}{y^{-6}z^3}$ $\frac{7x^4y^6}{z^3}$ ✓

3. Simplify the following expressions:

[6 marks]

- a) $5c - 7c$ $-2c$ ✓
 b) $4a - 5 + 7a$ $11a - 5$ ✓
 c) $3x^2 \times 5x$ $15x^3$ ✓
 d) $12p \div 4$ $3p$ ✓
 e) $\frac{30x^5y^6z}{24x^3y^3}$ $\frac{5x^2y^3z}{2}$ ✓✓

4. Expand the following:

[2 marks]

- a) $5(x + 3)$ $5x + 15$ ✓
 b) $2y(5 - y)$ $10y - 2y^2$ ✓

5. By expanding and then collecting like terms simplify

[2 marks]

$$3(p + 7) - 5(p - 4) = 3p + 21 - 5p + 20 = -2p + 41$$

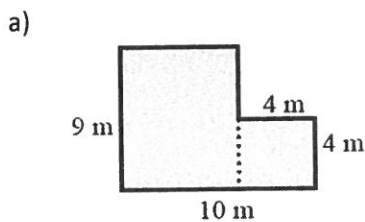
6. Fully factorise the following expressions

[4 marks]

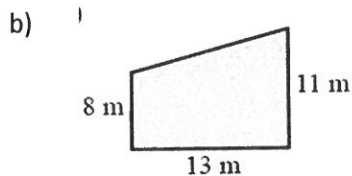
- a) $5t + 10$ $5(t + 2)$ ✓✓
 b) $9h^2 - 6h$ $3h(3h - 2)$ ✓✓

7. Calculate the area of the following shapes

[4 marks]



$$4 \times 4 + 9 \times 6 = 70 \text{ m}^2$$

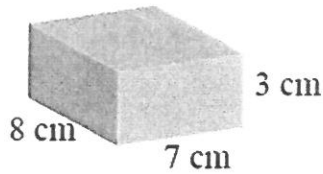


$$\begin{aligned} & 13 \times 8 + \frac{1}{2} \times 13 \times 3 \\ & = 104 + 19.5 \\ & = 123.5 \text{ m}^2 \end{aligned}$$

8. Calculate the Total surface area of the following shapes

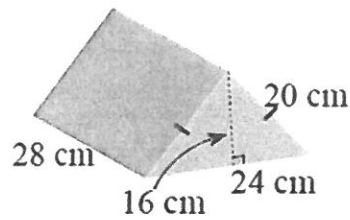
[6 marks]

i)



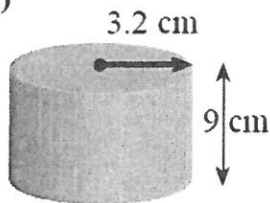
$$2 \times (8 \times 7 + 8 \times 3 + 7 \times 3) \\ = 2 \times (56 + 24 + 21) \\ = 182 \text{ cm}^2$$

ii)



$$(20 + 20 + 24) \times 28 + (16 \times 24)$$

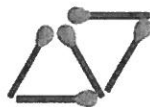
iii)



$$2\pi 3.2^2 + 9 \times 2 \times 3.2\pi$$

9. Write the rule for the following pattern where s is the shape and m is the number of matches.

[1 mark]



$$m = 2s + 1$$

10. Write the rule for the following tables:

[4 marks]

a)

x	0	1	2	3	4
y	5	8	11	14	17

$$y = 3x + 5$$

b)

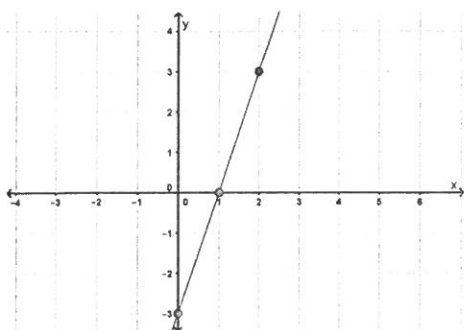
x	0	1	2	3	4
y	10	8	6	4	2

$$y = -2x + 10$$

11. Write the rule for the following graphs

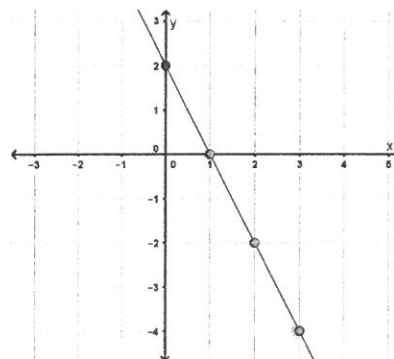
[4 marks]

a)



$$y = 3x - 3$$

b)

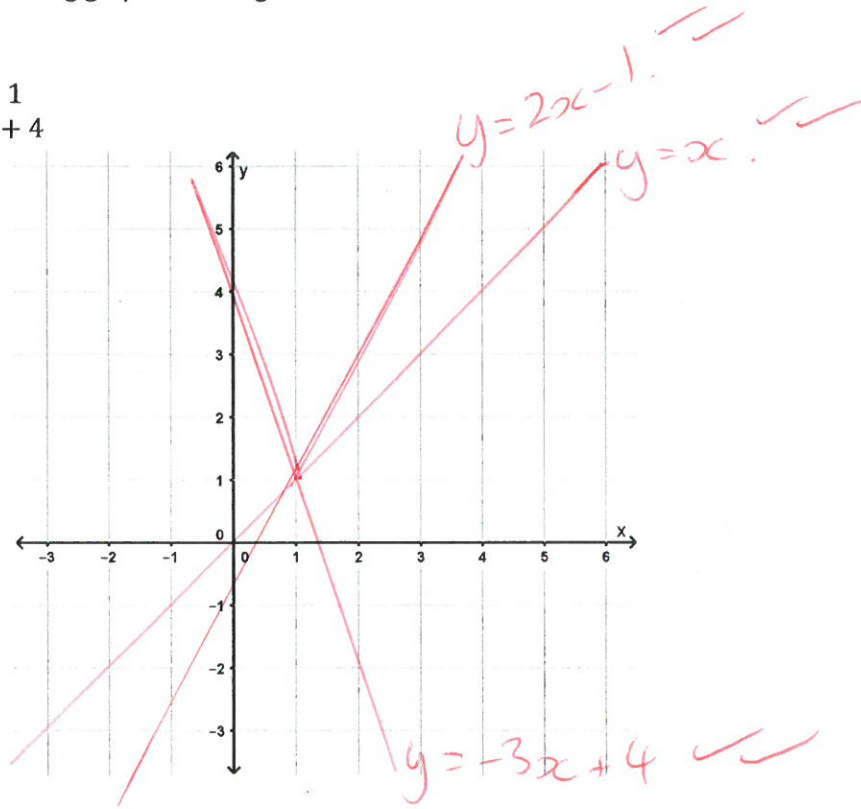


$$y = -2x + 2$$

12. Sketch the following graphs on the grid below

[6 marks]

- a) $y = x$
- b) $y = 2x - 1$
- c) $y = -3x + 4$



End of Paper.